Utku Kalkanlı

Munich, Germany · +49 172 7948165 · kalkanliutku@gmail.com · LinkedIn · GitHub · Website

PROFESSIONAL SUMMARY

Machine Learning Engineer with hands-on experience building end-to-end computer vision, MLOps, and agentic AI systems for industrial applications. Currently completing a Master's thesis at Fraunhofer IKS on zero-shot 3D vision-based anomaly detection, comparing feed-forward 3D reconstructions against CAD models. At Linde GmbH, deployed an RF-DETR detector that lifted small-object detection mAP from 80% to 95%, engineered the supporting Airflow/TensorRT pipelines, and is a named inventor of an AI-based fault classification method formally claimed by Linde under the German Employee Inventions Act. Proven capability navigating the full stack from advanced research to production deployment.

SKILLS & TECHNOLOGIES

Languages & Tools: Python · Linux · Git · Docker · FastAPI
ML Frameworks: PyTorch · TensorFlow · ONNX · TensorRT · Triton · CUDA
MLOps & Cloud: Microsoft Azure · Apache Airflow · MLflow · Kubeflow · Ray · Slurm
3D Vision: VGGT · DUSt3R · Open3D
NLP & LLMs: LangChain · LlamaIndex · RAG · Agentic AI
Methodologies: Agile · Scrum · CI/CD

EXPERIENCE

Computer Vision Research Assistant – **Fraunhofer IKS** Oct 2025 – Present | Munich, Germany

- Conducting Master's thesis research focused on 3D vision-based anomaly detection for industrial manufacturing objects.
- Developing a novel Zero-Shot Learning anomaly detection model, leveraging foundation models such as CLIP, DINOv2, and VGGT.
- Designing a VLM-based explainability stage that renders detected 3D anomalies into 2D views and generates natural-language defect descriptions for quality engineers.
- Building an agentic framework using LangGraph and Qwen tool-calling to orchestrate PyTorch-based domain adaptation experiments across diverse image domains.
- Reconstructing 3D representations from 2D object scenes using feed-forward reconstruction models to perform CAD-to-mesh anomaly comparison.

Machine Learning Engineer – **Linde GmbH – Digitalization Team** Oct 2023 – Present | Munich, Germany

- Optimized an existing Azure ML MLOps pipeline for the AIRGENIOUS platform, reducing ETL processing runtime by 37.5% (from 8 to 5 hours).
- Contributed to the system design and technical documentation of the AIRGENIOUS platform for MDR certification.
- Trained and deployed an RF-DETR model for satellite imagery small-object detection, boosting mAP from 80% to 95% and fully automating an industrial asset monitoring process; engineered an end-to-end MLOps pipeline using Apache Airflow for data ingestion, automated retraining, and TensorRT inference acceleration.
- Researched and developed CNNs and autoencoders in PyTorch for predictive maintenance, leveraging latent-space representations for out-of-distribution (OOD) anomaly detection.
- Co-authored an internal whitepaper; named inventor of "Orbit AI prediction and fault classification for Online Vibration Monitoring of critical rotating machinery" (invention P40700, formally claimed by Linde under the German Employee Inventions Act with inventor remuneration).

Interdisciplinary Project Student – **Covadonga Maritime** Nov 2024 – May 2025 | Remote

- Developed deterministic and deep learning methods for ship identification and tracking utilizing passive radar signals.
- Extracted signal features via FFT, spectrograms, and spectral peak detection; trained multivariate Bidirectional LSTM and CNN architectures in PyTorch.
- Achieved 91% classification accuracy on radar signal identification, significantly exceeding project benchmarks.

Machine Learning Engineering Intern – **Novit.AI** Mar 2022 – Jun 2022 | Ankara, Türkiye

- Engineered an end-to-end IoT application combining edge deployment on Raspberry Pi, a FastAPI backend, and a Vue.js frontend containerized via Docker.
- Trained an XGBoost prediction model; implemented automated CI/CD pipelines via GitHub Actions and configured NGINX for load balancing.

Earlier Internships – **Open Business Software Solutions · Cybersoft** 2020 – 2021 | Türkiye

- Full-stack software development internships: enterprise CRUD application (React, Django, PostgreSQL, Elasticsearch) and responsive MERN-stack web application.

OPEN SOURCE PROJECTS

Technical Lead & Top Collaborator – **Omdena Project** Oct 2022 – Mar 2023 | Remote

- Led a team of 4 to build an AI-generated closed captioning system supporting deaf and hard-of-hearing individuals.
- Implemented one-shot learning for actor detection; integrated OpenAI Whisper API and Hugging Face Transformers for dense video captioning.
- Awarded Technical Lead & Top Collaborator Certificate.

EDUCATION

M.Sc. Informatics – **Technical University of Munich (TUM)** Apr 2023 – Expected Oct 2026 | Munich, Germany

Major: Machine Learning & Analytics (Deep Learning, NLP, AI in Medicine, Data Visualization) · Minor: Scientific Computing & High-Performance Computing (HPC) · Engineering Software-Intensive Systems (SE) · Databases and Information Systems (DBI)

- Master's Thesis (with Fraunhofer IKS): *Zero-Shot 3D Vision-Based Anomaly Detection for Industrial Parts* – manuscripts in preparation for 3DV and ICRA.
- Engineering Praktikum: Implemented a coverage-guided fuzz testing framework to evaluate deep learning model robustness (MNIST, SVHN, GTSRB).

B.Sc. Computer Science – **Bilkent University** Sep 2017 – Jun 2022 | Ankara, Türkiye

Electives: Software Verification & Validation, ML Intro, Bioinformatics Algorithms, Machine-to-Machine Systems

- Senior Project: Android app for fatigue detection and facial expression recognition using OpenCV and a Flask backend hosted on AWS.

EXTRACURRICULAR & CERTIFICATIONS

Google Developers & Inzva ML Bootcamp · Jul – Nov 2022

- Completed 5-month TensorFlow & ML program; mentored fellow participants.
- Participated in inzva Deep Learning Study Group #8 (12-week deeplearning.ai curriculum).

Certifications

- Deep Learning Specialization – DeepLearning.AI / Coursera (Nov 2022)
- TensorFlow Developer Specialization – DeepLearning.AI / Coursera (Nov 2022)

LANGUAGES

Turkish (Native) · **English** (Professional Working Proficiency) · **German** (Intermediate – B1)